Salt, Fat, Acid, Distance

Seeing how culinary resemblance exceeds what geographic distance predicts

Matthew Scott Tan

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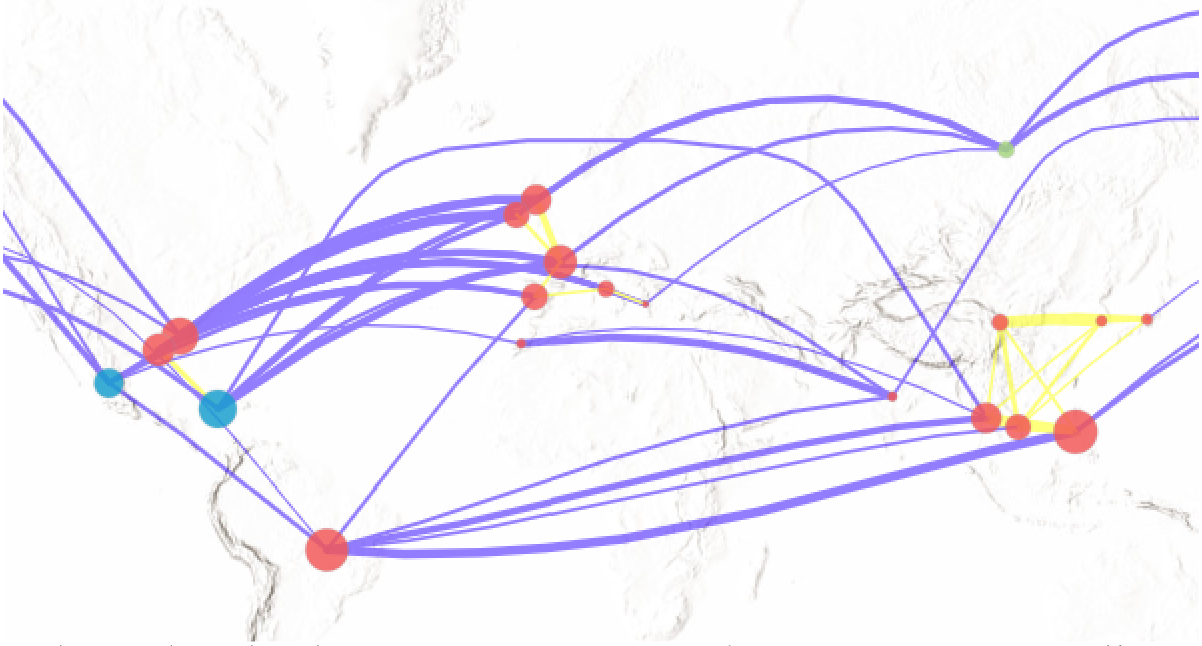
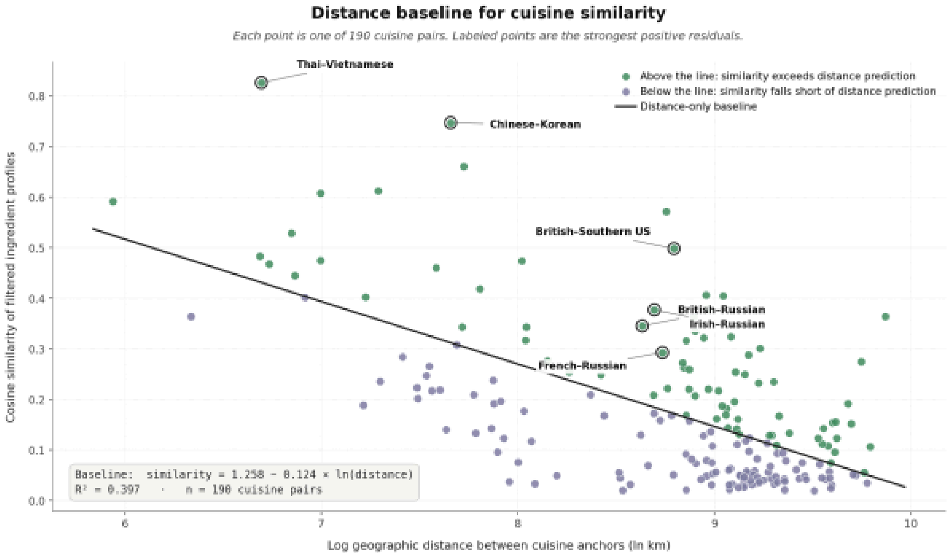
Cuisines are geographic: crops, climates, and terrain shape what people cook. But cuisines also travel through migration, trade, and colonial exchange. This project asks: after geographic distance is accounted for, which cuisines remain unexpectedly similar? I compare 20 cuisine-labeled recipe profiles, anchor each cuisine geographically, compute 190 great-circle distances, and model how ingredient similarity declines with distance. The residuals, observed similarity minus distance-predicted similarity, become the map. Positive residuals reveal where cuisines are more alike than proximity predicts, pointing to archipelagos, peninsulas, Atlantic shores, and long-range exchange as spatial structures that distance alone cannot explain.



Cuisines are connected here by ingredient resemblance that distance alone does not explain. Blue links mark the East and Southeast Asia focused case, while orange links mark long-distance positive residuals. The map shows candidate spatial associations rather than proven routes of exchange.

Turning Cuisine Similarity into a Map

The GIS workflow turns cuisine labels into spatial evidence: anchor each cuisine, compute 190 great-circle pair distances, fit a log-distance similarity baseline, map residual corridors, then test the residual field with Mantel statistics, Local Moran's I, and a reproducible bridge index. Distance still matters: similarity declines as log distance increases (slope -0.124, $R^2 = 0.397$). But roughly 60 percent of observed variation remains outside the distance baseline, which is why residual mapping is informative.



The regression line is the distance baseline. Points sitting above it are cuisine pairs more similar than distance predicts, and those positive residuals become the mapped network analyzed in the rest of the StoryMap.

Dataset	Source	Spatial unit
Yummly recipe corpus	Yummly/Kaggle 2015 [1]; prepared by D. Zelený 2024 [2]	39,774 recipes; 20 cuisines
Cuisine anchors	Project-curated	20 lat/lon centroids
Pairwise distance	GeoPy great-circle	190 pairs, WGS84
Natural Earth basemap	Natural Earth v5.1.1	110m global / 50m regional
UN M49 subregions	UN Statistics Division	Country classification
Colonial-administration crosswalk	Project-internal	190 pairs, 3-tier ordinal

Use in this project
Cuisine-by-ingredient similarity matrix
Distance and mapping

Use in this project
Distance baseline regression
Cartographic context
Same-subregion adjacency control
Colonial partial-Mantel covariate

The colonial-administration crosswalk codes each of the 190 cuisine pairs on a 3-tier ordinal scale: 0 = no shared colonial-administration history; 1 = indirect or partial overlap, such as adjacent colonies of the same power or one cuisine's region falling within the metropole of the other; 2 = direct shared administration, where both regions were administered by the same colonial power, or one was administered by the polity associated with the other, within roughly 1500–1960. Coding follows standard references for the Spanish, Portuguese, British, French, Dutch, and U.S. colonial empires. The variable is deliberately coarse — it cannot resolve duration, intensity, or institutional form of contact — and enters the analysis only as a partial-Mantel covariate, not as a causal claim about exchange routes.

The Residual Map Has Spatial Structure

Three independent tests confirm the network has statistically significant spatial structure. A Mantel test on all 190 cuisine pairs finds a strong distance-dissimilarity relationship that survives partialling out same-subregion adjacency, indicating the signal is not merely a neighbors-are-neighbors artifact. Local Moran's I then locates roles in the residual field, identifying Russian as a significant low-low outlier while Mexican and Jamaican form a high-high Atlantic-rim cluster. A partial Mantel test detects a modest but robust correlation between residual similarity and shared colonial-administration history, even after controlling for both distance and adjacency.

The residual network is statistically spatial. Mantel tests validate the distance relationship, while Local Moran's I identifies cuisines whose residual position differs from or clusters with nearby anchors.

All three statistical layers appear here in one view. The Mantel r is $+0.630$, with partial r of $+0.512$ after controlling for subregion. The Russian LL classification carries $p = 0.009$, and the colonial partial Mantel r is $+0.181$ at $p = 0.022$ after applying the full set of controls.

Bridge Cuisines Reveal the Network’s Structure

Aggregating pairwise residuals to a cuisine-level network position reveals that the residual geography is anchored by a small set of "bridge" cuisines rather than by a single region. The strongest bridge scores belong to Filipino at 0.79, Southern U.S. at 0.76, French at 0.74, Cajun-Creole at 0.73, Brazilian at 0.66, and Thai at 0.65, occupying geographically distinct positions across Pacific-archipelagic, Atlantic-rim, European-peninsular, Gulf-Caribbean, and mainland Southeast Asian space. The bridge index itself is a five-component score, fully reproducible from the shipped reference inputs at seed = 42.

Bridge cuisines occupy different geographies rather than clustering in one region. The highest scores identify the nodes connecting otherwise separated parts of the residual network.

Robustness checks support the cluster-level pattern: Filipino remains a frequent top bridge in resampling, and Russia's low-low LISA role is directionally stable across anchor choices.

Where Distance Fails: East and Southeast Asia

The clearest regional corridor is East and Southeast Asia. Chinese-Korean is the largest positive residual in the corpus at +0.44, Thai-Vietnamese is second at +0.40, and Filipino-Thai (+0.36), Filipino-Vietnamese (+0.25), and Chinese-Japanese (+0.21) form a compact corridor linking mainland adjacency, island geography, and archipelagic extension. This is the easiest place to see why the method matters, because the map does not merely label similar cuisines, it shows where distance explains similarity and where it fails.

East and Southeast Asia provide the residual method with its cleanest regional example. The strongest links combine nearby mainland pairs with archipelagic extensions that distance alone underpredicts.

Spotlight: Filipino Cuisine

Filipino is the strongest single bridge in the network. Its top residual partners combine regional and transoceanic ties, with Thai (+0.36), Brazilian (+0.32), Vietnamese (+0.25), Jamaican (+0.13), and Southern U.S. (+0.07) appearing among them, and the mixture is the project's most interpretable hypothesis generator. The same cuisine connects the Southeast Asian corridor to Atlantic and Gulf cuisines in a pattern consistent with Spanish colonial and maritime exchange, an interpretation the colonial partial Mantel result above supports in modest form.

Filipino is the highest-scoring bridge cuisine. Its residual partners combine nearby Southeast Asian links with long-distance Atlantic and Gulf links, making it the clearest single anchor in the network.

What the Map Reveals

Three structural patterns emerge from the residual surface. The largest unexplained similarities are regional corridors close in: Chinese-Korean (+0.44), Thai-Vietnamese (+0.40), and the Filipino-Thai-Vietnamese-Chinese-Japanese cluster sit furthest above the distance line, combining mainland adjacency with archipelagic extension that great-circle distance treats as remote. Second, the residual network is anchored not by a single region but by a geographically diverse set of bridge cuisines — Filipino, Southern U.S., French, Cajun-Creole, Brazilian, and Thai — connecting otherwise separated parts of the network from Pacific-archipelagic, Atlantic-rim, European-peninsular, Gulf-Caribbean, and mainland Southeast Asian positions. Third, after controlling for both distance and same-subregion adjacency, shared colonial-administration history still leaves a modest but robust trace (partial Mantel $r = +0.181$, $p = 0.022$); Filipino's combination of nearby Southeast Asian links with long-distance Atlantic and Gulf ties is the clearest single expression of that signal.

These are exploratory findings on a 20-cuisine corpus, not a complete map of world food. Cuisine anchors are approximate centroids, the recipe corpus is platform-mediated and regionally uneven, and the residuals identify candidate spatial associations rather than proven routes of exchange. What the map contributes is a structured account of where geography stops explaining: archipelagos, peninsulas, Atlantic shores, and colonial corridors emerge as a patterned residual geography that distance alone collapses into noise. Distance explains part of cuisine resemblance; GIS reveals the shape of what distance leaves behind — and that shape is structured, not random.

> *Cuisine resemblance is not only about who lives nearby. It is also about who has been connected—and those connections have a shape a map can show.*

References

[1] Yummly. (2015). What's Cooking. Kaggle dataset.
[2] Zelený, D. (2024). Recipes dataset, anadat-r.
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ChatGPT was used to help with image and figure generation